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Hearing device chargers

Abstract

A hearing device charger including a charger housing and a hearing device support post, associated with the charger housing, including a top wall, with a top recess that extends a first distance in a first direction from an open end to a closed end and is configured to receive a portion of the hearing device housing, and a side wall that extends away from the top wall in the first direction and defines a support post outer perimeter and a support post outer recess that extends inwardly from the support post outer perimeter and extends a second distance away from the top wall in the first direction that is greater than the first distance and is configured to receive the hearing device removal handle. A transmitter coil may be associated with the hearing device support post, and a power supply may be operably connected to the transmitter coil.

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Background/Summary

BACKGROUND

1. Field

(1) The present disclosure relates generally to hearing device chargers.

2. Description of the Related Art

(2) In-the-ear hearing devices, in-the-canal hearing devices and completely-in-the-canal hear devices (collectively referred to herein as “ITE hearing devices”) may include a housing that has a main portion with a sound output port that is positioned facing to the tympanic membrane and a faceplate at the lateral end of the main portion. Hearing device components, such as a microphone, a receiver, electronics and a rechargeable battery (e.g., a rechargeable lithium-ion battery) may be located with the housing main portion and/or faceplate. Ambient sound pressure waves are picked up by the microphone and converted into electrical signals. The electrical signals, in turn, are processed by sound processor circuitry. The processed signals drive the receiver, which delivers amplified (or otherwise processed) sound pressure waves to the ear canal.

(3) The ITE hearing device battery may in some instances be wirelessly recharged with a battery charger. Here, the ITE hearing device includes a charging receiver coil and the battery charger includes a charging transmitter coil. The transmitter coil may be carried within a charging post on which the ITE hearing device is mounted during charging, and the charging post and ITE hearing device may be configured in such a manner that an optimal coil alignment is maintained. The charging post and ITE hearing device may also include magnets that secure the ITE hearing device to the charging post. Exemplary chargers are illustrated and described in US Pat. Pub. Nos. 2008/0118093A1 and 2021/0385591A1.

(4) The present inventors have determined that the battery chargers associated with ITE hearing devices are susceptible to improvement. For example, the present inventors have determined that the charging posts in conventional battery chargers are not able to accommodate the removal handles that extend laterally from the faceplate of some ITE hearing devices.

SUMMARY

(5) A hearing device charger in accordance with at least one of the present inventions comprises a charger housing and a hearing device support post, associated with the charger housing, including a top wall, with a top recess that extends a first distance in a first direction from an open end to a closed end and is configured to receive a portion of the hearing device housing, and a side wall that extends away from the top wall in the first direction and defines a support post outer perimeter and a support post outer recess that extends inwardly from the support post outer perimeter and extends a second distance away from the top wall in the first direction that is greater than the first distance and is configured to receive the hearing device removal handle. A transmitter coil may be associate with the hearing device support post, and a power supply may be operably connected to the transmitter coil.

(6) A hearing system in accordance with at least one of the present inventions comprises an in-the-ear (“ITE”) hearing device and a hearing device charger. The hearing device may include an ITE housing defining a medial end and a lateral end, a microphone, a receiver, a rechargeable battery, a

hearing device magnet and a receiver coil associated with the housing, and a removal handle extending laterally from the lateral end of the housing. The hearing device charger may include a charger housing, a hearing device support post, associated with the charger housing and including a top wall and a support post outer recess, a magnet associated with the hearing device support post, a transmitter coil associated with the hearing device support post, and a power supply operably connected to the transmitter coil. The hearing device magnet and the support post magnet may be respectively oriented such that they are attracted to one another when the lateral end of the ITE housing abuts the top wall of the hearing device support post, and the respective configurations of the ITE hearing device and the hearing device support post may be such that the removal handle is located within the support post outer recess when the lateral end of the ITE housing abuts the top wall of the hearing device support post.

(7) There are a variety of advantages associated with such chargers and systems. By way of example, but not limitation, the support posts allow hearing devices to be placed onto the charger in the intended manner despite the presence of removal handles, thereby facilitating the charging of hearing devices with removal handles.

(8) The above described and many other features of the present inventions will become apparent as the inventions become better understood by reference to the following detailed description when considered in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Detailed descriptions of the exemplary embodiments will be made with reference to the accompanying drawings.

(2) FIG. 1 is a perspective view of a hearing device charger in accordance with one embodiment of a present invention.

(3) FIG. 2 is a front view of a portion of the hearing device charger illustrated in FIG. 1.

(4) FIG. 3 is a top view of a portion of the hearing device charger illustrated in FIG. 1.

(5) FIG. 4 is a perspective view of an exemplary hearing device that may be used in conjunction with the hearing device charger illustrated in FIG. 1.

(6) FIG. 5 is a functional block diagram of the hearing device charger illustrated in FIG. 1.

(7) FIG. 6 is a side view of the hearing device illustrated in FIG. 4.

(8) FIG. 7 is a perspective view of the hearing device illustrated in FIG. 4.

(9) FIG. 8 is a perspective view of a portion of the hearing device illustrated in FIG. 4.

(10) FIG. 9 is a functional block diagram of the hearing device illustrated in FIG. 4.

(11) FIG. 10 is a perspective view of a portion of the hearing device charger illustrated in FIG. 1.

(12) FIG. 11 is a top view of a portion of the hearing device charger illustrated in FIG. 1.

(13) FIG. 12 is a front view of a portion of the hearing device charger illustrated in FIG. 1.

(14) FIG. 13 is an exploded perspective view of a portion of the hearing device charger illustrated in FIG. 1.

(15) FIG. 14 is a perspective view of the hearing device illustrated in FIG. 4 and a portion of the hearing device charger illustrated in FIG. 1 in a disengaged state.

(16) FIG. 15 is a perspective view of the hearing device illustrated in FIG. 4 and a portion of the hearing device charger illustrated in FIG. 1 in an engaged state.

(17) FIG. 16 is a side view of a hearing device and a charging surface of a conventional charger.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

(18) The following is a detailed description of the best presently known modes of carrying out the inventions. This description is not to be taken in a limiting sense, but is made merely for the purpose of illustrating the general principles of the inventions. The present inventions also have

application in a wide variety of hearing devices that provide sound (i.e., either sound or a perception of sound) to the hearing impaired as well as others who require such hearing devices on a situational basis. Examples of such hearing devices include hearing aids, such as the ITE hearing aids described herein. The present inventions are not, however, limited to ITE hearing aids, and may be employed in combination with other hearing devices that currently exist, or are yet to be developed.

(19) It should also be noted that if and when used herein, the term “lateral” refers to the direction and parts of hearing devices which face away from the tympanic membrane, the term “medial” refers to the direction and parts of hearing devices which face toward the tympanic membrane, the term “superior” refers to the direction and parts of hearing devices which face the top of the head, the term “inferior” refers to the direction and parts of hearing devices which face the feet, the term “anterior” refers to the direction and parts of hearing devices which face the front of the body, and the “posterior” refers to the direction and parts of hearing devices which face the rear of the body.

(20) Referring to FIGS. 1-3, the exemplary hearing device charger (or “charger”) **100** includes a base **102**, with a housing **104** in which various components are located and on which various components are supported, and a cover **106** that may be pivotably mounted to the base **102**. The exemplary housing **104** has a bottom wall **108**, a perimeter wall **110**, and an interior wall **112**. A power port **114** extends through the perimeter wall **110** and a visible indicator **116**, such as the illustrated set of LEDs, is associated with the perimeter wall. The cover **106**, which is shown in a closed state in FIG. 1 and is omitted from FIGS. 2 and 3, may be pivotably secured to the base **102** with a hinge pin **118** and may be maintained in a closed state with a latch **120** on the housing perimeter wall **110** and an indentation (not shown) on the cover **106**. The exemplary charger **100** is configured to accommodate hearing devices, such as the hearing device **200** illustrated in FIG. 4, that includes a removal handle, such as the removal handle (or “handle”) **232**. To that end, the charger **100** includes a pair of hearing device support posts **122** that are configured to accommodate handle **232** of the hearing device **200** as well as other handles of other hearing devices. The support posts **122** may be mounted on, or may extend through, the interior wall **112**. Suitable materials for the base **102** and cover **106** include but are not limited to, polycarbonate/acrylonitrile butadiene styrene (PC/ABS) such as Sabic CYCOLY C1200HF or ABS such as Chimei POLYLAC PA757F.

(21) The exemplary hearing device **200** is described in greater detail below with reference to FIGS. 6-9, the exemplary hearing device support posts **122** are described in greater detail below with reference to FIGS. 10-13, and the manner in which the handle **232** is accommodated by the hearing device support posts **122** is described in greater detail below with reference to FIGS. 14 and 15.

(22) Although the present inventions are not so limited, the exemplary hearing device charger **100** is configured to wirelessly charge hearing devices such as the hearing device **200**. To that end, and referring to FIG. 5, an antenna coil **124** that transmits power to a corresponding hearing device antenna coil **216** (FIGS. 8 and 9) via inductive coupling and magnets **126** that are attracted to hearing device magnets **218** (FIGS. 8 and 9) may be located within each of the support posts **122**. Transmitter circuits (or “charge circuits”) **128** drive the antenna coils **124**. The magnets **126** and **218** secure the hearing device **200** to the charging post **122**. Power may be provided by a power supply **130** (e.g., an external USB power supply) that may be connected to the power port **114** as well as by a rechargeable battery (or other energy storage device) **132**. Operation of the charger is controlled by control circuitry **134**.

(23) As illustrated for example in FIGS. 6-9, the ITE hearing device **200** includes a housing **202**, with a shell **204** that defines a shape corresponding to the ear canal and a faceplate **206** mounted to the lateral end of the shell, as well as components such as a microphone **208**, a receiver **210**, a rechargeable battery or other rechargeable power source **212**, and sound processing electronics **214** within the housing **202**. An antenna coil **216** that receives power from one of the charger antenna coils **124** (FIGS. 5 and 13) via inductive coupling is located within the faceplate **206**, as are

magnets **218** that are attracted to the charger magnets **126** (FIGS. **5** and **13**). For example, the magnets **126** and **218** define respective N-poles and S-pole and the magnets may be oriented in such a manner that the N-poles of the magnets **126** and face the S-poles of the magnets **218**. The housing shell **204** is sized and shaped for positioning within the ear canal and includes a sound output aperture **220** and a venting aperture **222**. The shape of the housing shell **204** may be a generic shape that is suitable for a large number of patients or may be a custom shape that is 3D printed or otherwise formed for the ear canal of a particular patient. Suitable housing materials include, but are not limited to, plastics such as an acrylic and metals such as titanium.

(24) Referring more specifically to FIGS. **6** and **7**, the exemplary faceplate **206** includes a main portion **224** and a projection **226**, with a push button **228** that may perform various functions and a sound input aperture **230**, that extends laterally from the main portion. The projection **226** is configured, i.e., is sized and shaped, to be received by a support post top recess **138** (FIGS. **10** and **11**) in such a manner that the antenna coil **216** will be properly aligned with the charger antenna coil **124** and the magnets **218** will be properly aligned with the charger magnets **126**. A handle **232**, which may be used to remove the hearing device from the recipient's ear, extends laterally from the faceplate main portion **224**.

(25) Turning to FIGS. **10-13**, the exemplary hearing device support posts **122** each include a top wall **136** with a top wall recess (or “recess”) **138**, that extends from an open end **140** to a closed end **142**, and a side wall **144** that extends away from the top wall to a bottom wall **146**. The side wall **144** extends around the top and bottom walls **136** and **146** and defines an outer perimeter P. Although the number may be increased or decreased on other implementations, the side wall **144** defines four outer recesses **148-1** to **148-4** (collectively outer recesses **148**) that extend inwardly from the outer perimeter P. The outer recesses **148** extend the entire length L of the support posts **122**, from the top wall **136** to the bottom wall **146**, in the illustrated implementation. The outer recesses **148** extend less than the entire length L of the support post, i.e., from the top wall **136** to a point between the top wall and the bottom wall **146**, in other implementations. In either case, the outer recesses **148** should be as long or longer than the associated hearing device handle **232**. The length of the recesses **148** may in some instances range from 7.00 mm to 13.00 mm, which will accommodate most handles.

(26) The exemplary top wall recess **138** is configured, i.e., is sized and shaped, to receive the associated hearing device faceplate projection **226**. In the illustrated implementation, the recess **138** and faceplate projection **226** are mechanically keyed so that there are two possible orientations of the faceplate projection relative to the recess and, therefore, two possible orientations of the hearing device **200** relative to the associated support post **122**. The orientations are offset by 180 degrees. By way of example, by not limitation, the recess **138** and faceplate projection **226** are generally rectangular in shape with curved longitudinal ends and straight sides. The recess **138** may also be configured to accommodate the button **228** on the hearing device faceplate projection **226** and, for example, may include a pair of button recesses **150** that are located and configured, i.e., are sized and shaped, such that the button **228** will be received in one other recesses when the hearing device faceplate projection **226** is located within the recess **138**.

(27) The exemplary support post **122** illustrated in FIGS. **10-13** has a cruciform shape when viewed from the above or in a plane perpendicular to the length L. Other shapes, including but not limited to shapes that define non-circular outer perimeters, may be employed. The number and location of the recesses may be adjusted to accommodate various removal handles and removal handle locations.

(28) The supports posts may be formed and assembled in any suitable fashion. As illustrated in FIG. **13**, the exemplary support posts **122** include a base **152**, on which the antenna coil **124** and magnets **126** are supported, and a cover **154**. The base **152** includes the bottom wall **146** and portions of the side wall **144**, while the cover **154** includes the top wall **136** and the remaining portions of the side wall. The magnets **126** may be located within the base **102**

(29) As alluded to above, the present charging posts are configured to accommodate hearings devices with a laterally extending removal handles and, in particular, the charging posts includes the recesses that provide clearance for a handle. To that end, and referring to FIG. 14, the exemplary hearing device 200 may be mounted onto the support posts 122 of the exemplary charger 100 by first orienting the hearing device relative to a support post in such a manner that the faceplate projection 226 is aligned with the recess 138 and the handle 232 is aligned with one of the outer recesses 148 (here, outer recess 148-3). The hearing device 200 may then lowered onto the support post 122, until the faceplate projection 226 is nested within the recess 138 and the handle 232 is within the outer recess 148-3, thereby aligning the magnets 126 and 218 and the coils 124 and 216. In those instances where the hearing device 200 is rotationally offset by 180 degrees from the orientation illustrated in FIGS. 14 and 15, the handle 232 will be located in the outer recess 148-1 (FIGS. 10 and 11). The outer recess 148-2 and 148-4 may be used to accommodate handles located in other locations on the faceplate 206.

(30) By way of comparison, the handle 232 would interfere with placement of the hearing device 200 onto the charging surface CS of a conventional charger that lacks the present recesses, as shown in FIG. 16.

(31) Although the inventions disclosed herein have been described in terms of the preferred embodiments above, numerous modifications and/or additions to the above-described preferred embodiments would be readily apparent to one skilled in the art. By way of example, but not limitation, the inventions include any combination of the elements from the various species and embodiments disclosed in the specification that are not already described. It is intended that the scope of the present inventions extend to all such modifications and/or additions and that the scope of the present inventions is limited solely by the claims set forth below.

Claims

1. A hearing device charger for use with a hearing device the includes a housing and a removal handle that extends laterally from the housing, the hearing device charger comprising: a charger housing; a hearing device support post, associated with the charger housing, including a top wall, with a top wall recess that extends a first distance in a first direction from an open top end to a closed bottom end and is configured to receive a portion of the hearing device housing when the hearing device is mounted onto the hearing device support post, and a side wall that extends away from the top wall in the first direction and defines a support post outer perimeter and a support post outer recess on the side wall, wherein the support post outer recess is dimensioned and positioned relative to the top wall recess such that the hearing device removal handle will be within support post outer recess when the hearing device is mounted onto the hearing device support post; a transmitter coil associated with the hearing device support post; and a power supply operably connected to the transmitter coil.
2. A hearing device charger as claimed in claim 1, further comprising: a magnet located adjacent to or within the hearing device support post.
3. A hearing device charger as claimed in claim 1, wherein the top wall recess defines a non-circular shape in a plane perpendicular to the first direction.
4. A hearing device charger as claimed in claim 1, wherein the first direction is perpendicular to the top wall.
5. A hearing device charger as claimed in claim 1, wherein the transmitter coil is located within the hearing device support post below the top wall recess.
6. A hearing device charger for use with a hearing device the includes a housing and a removal handle that extends laterally from the housing, the hearing device charger comprising: a charger housing; a hearing device support post, associated with the charger housing, including a top wall, with a top wall recess that extends a first distance in a first direction from an open end to a closed

end and is configured to receive a portion of the hearing device housing, and a side wall that extends away from the top wall in the first direction and defines a support post outer perimeter and a plurality of outer recesses that extend inwardly from the support post outer perimeter and extend a second distance away from the top wall in the first direction that is greater than the first distance and are configured to receive the hearing device removal handle; a transmitter coil associated with the hearing device support post; and a power supply operably connected to the transmitter coil.

7. A hearing device charger as claimed in claim 6, wherein the hearing device support post includes four outer recesses.

8. A hearing device charger as claimed in claim 6, wherein the hearing device support post defines a cruciform shape in a plane perpendicular to the first direction.

9. A hearing device charger as claimed in claim 1, wherein the top wall recess includes first and second button recesses.

10. A hearing device charger as claimed in claim 1, wherein the charger housing includes a base and a cover that pivotably connected to the base and is movable between an open state and a closed state; and the hearing device support post is secured to the base.

11. A hearing system, comprising: an in-the-ear (“ITE”) hearing device including an ITE housing defining a medial end and a lateral end, a microphone, a receiver, a rechargeable battery, a hearing device magnet and a receiver coil associated with the housing, and a removal handle extending laterally from the lateral end of the housing; and a hearing device charger including a charger housing, a hearing device support post, associated with the charger housing, including a top wall, and a support post outer recess, a charger post magnet associated with the hearing device support post, a transmitter coil associated with the hearing device support post, and a power supply operably connected to the transmitter coil; wherein the hearing device magnet and the support post magnet are respectively oriented such that they are attracted to one another when the lateral end of the ITE housing abuts the top wall of the hearing device support post; and wherein the respective configurations of the ITE hearing device and the hearing device support post are such that the removal handle is located within the support post outer recess when the lateral end of the ITE housing abuts the top wall of the hearing device support post.

12. A hearing system as claimed in claim 11, wherein the hearing device support post includes a top wall recess that defines a size and a shape; and the ITE housing includes a faceplate with a projection that defines a size and a shape which correspond to the size and shape of the top wall recess.

13. A hearing system as claimed in claim 12, wherein the top wall recess shape is a non-circular shape; and the faceplate projection shape is the non-circular shape.

14. A hearing device charger as claimed in claim 11, wherein the hearing device support post defines a support post outer perimeter; and the support post outer recess extends inwardly from the support post outer perimeter.

15. A hearing device charger as claimed in claim 11, wherein the transmitter coil and the hearing device magnet are located within the support post.

16. A hearing device charger as claimed in claim 11, wherein the hearing device support post includes a plurality of outer recesses.

17. A hearing device charger as claimed in claim 11, wherein the hearing device support post includes four outer recesses.

18. A hearing device charger as claimed in claim 11, wherein the hearing device support post defines a cruciform shape in a plane perpendicular to the first direction.

19. A hearing device charger as claimed in claim 11, wherein the top wall recess includes first and second button recesses.

20. A hearing device charger as claimed in claim 11, wherein the charger housing includes a base and a cover that pivotably connected to the base and is movable between an open state and a closed state; and the hearing device support post is secured to the base.

